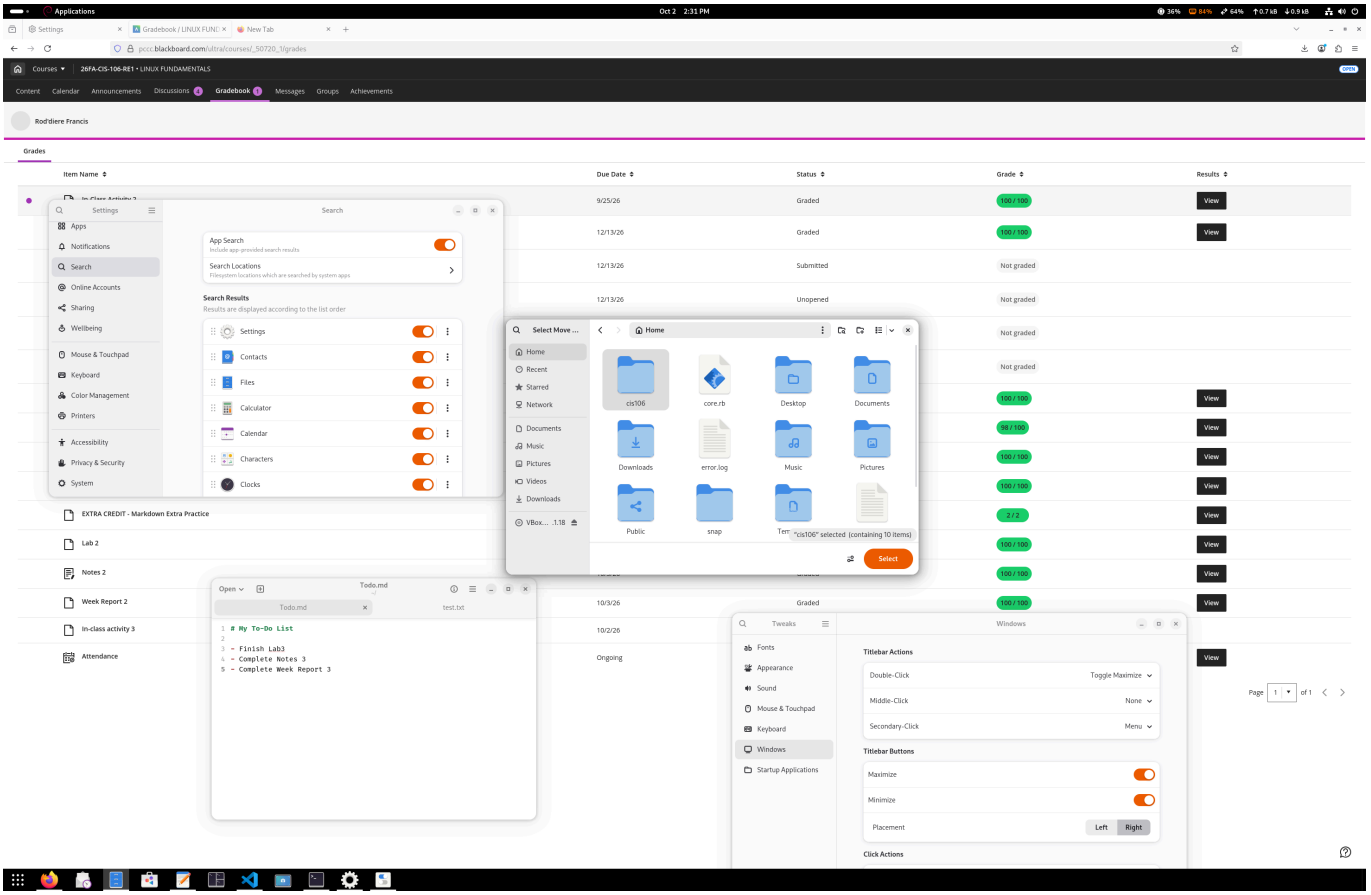
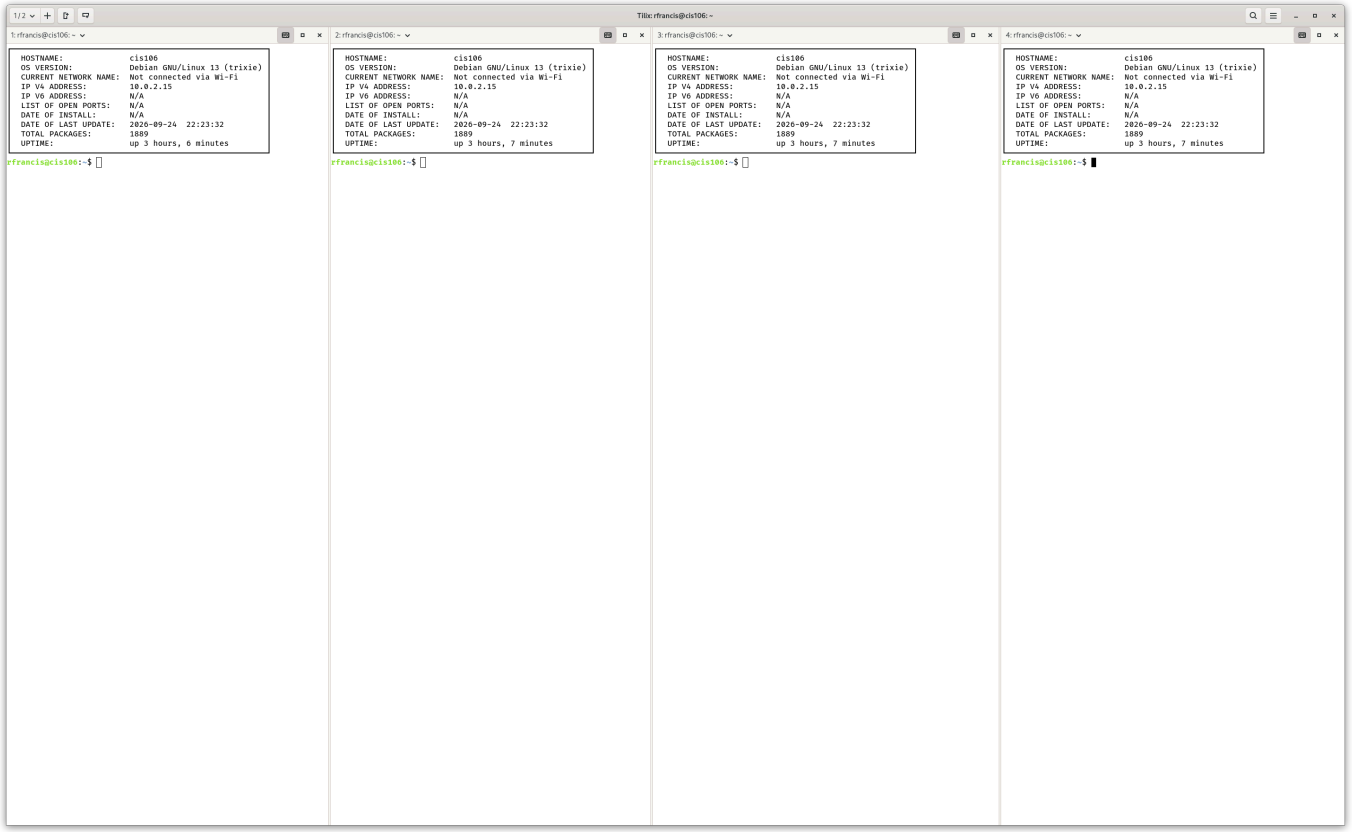
Lab 3 Submission

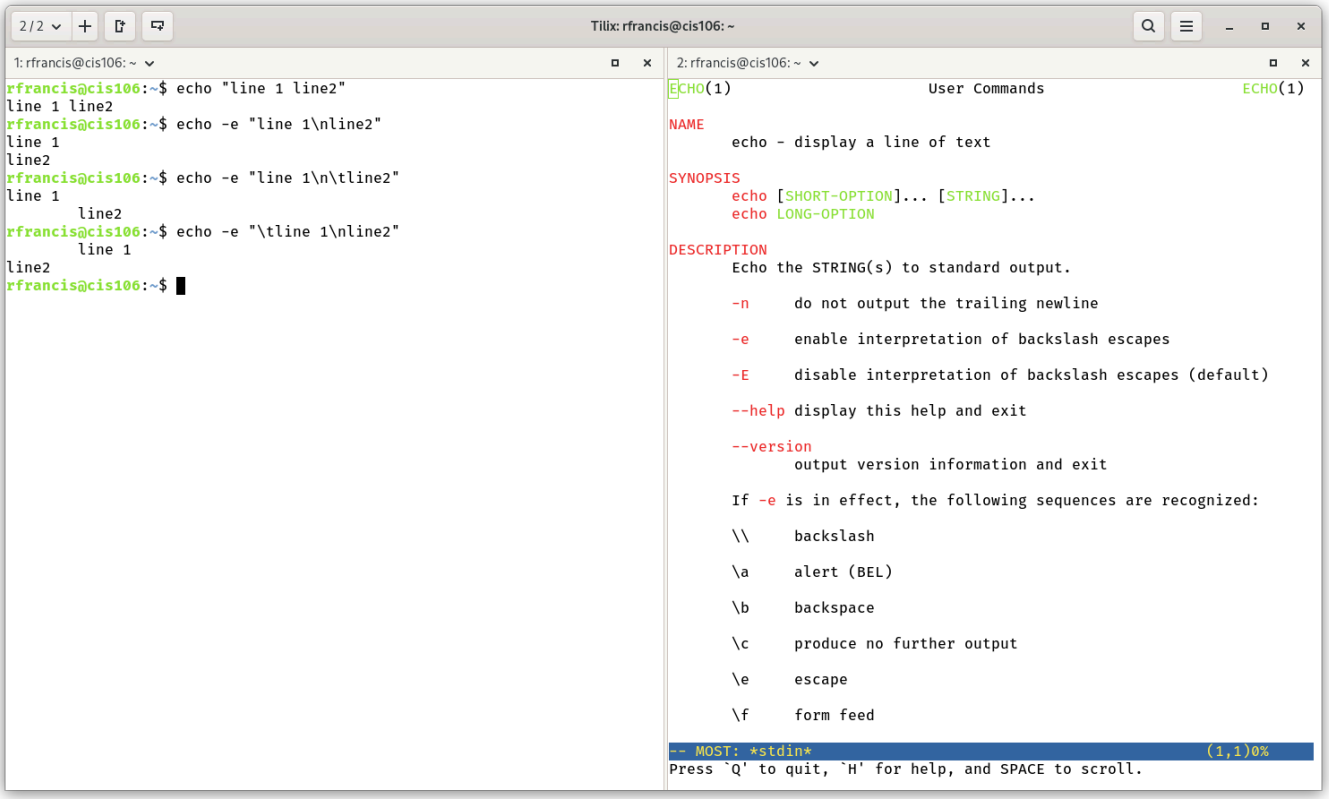
Question 1



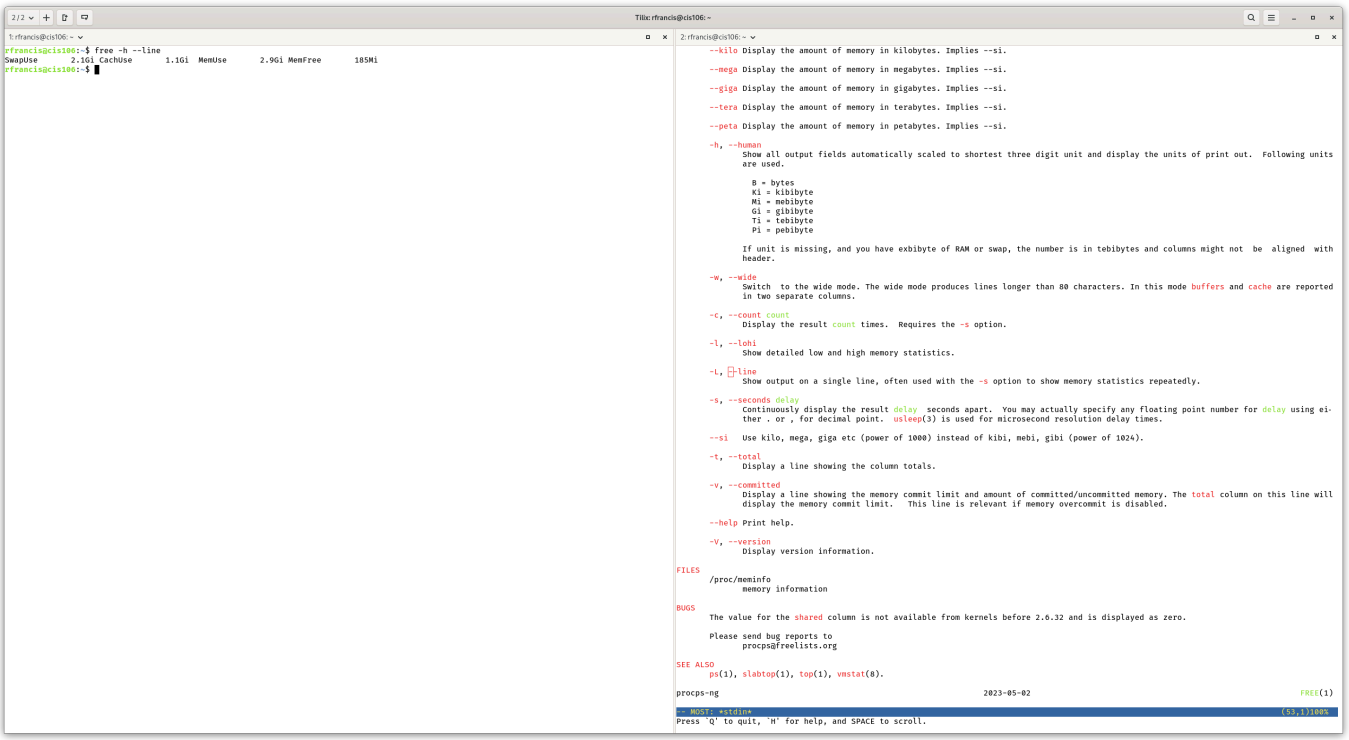
Question 2

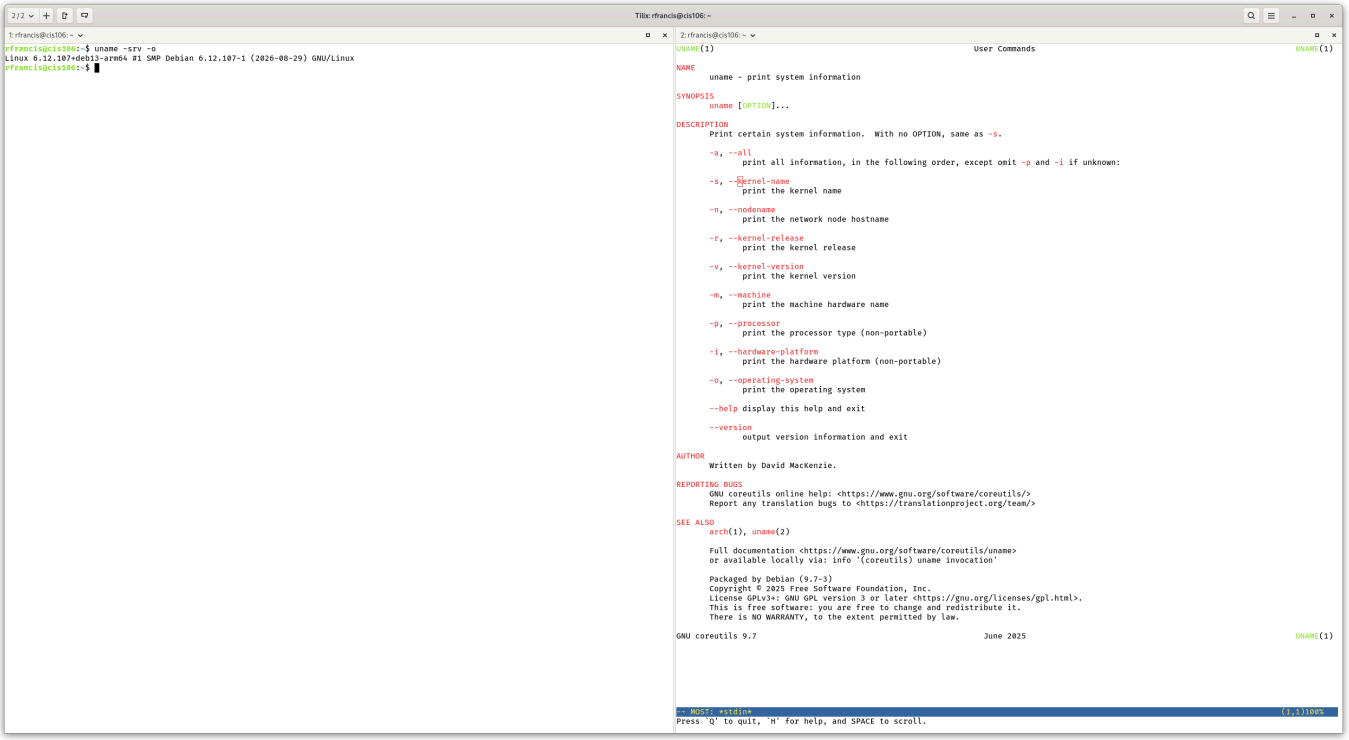


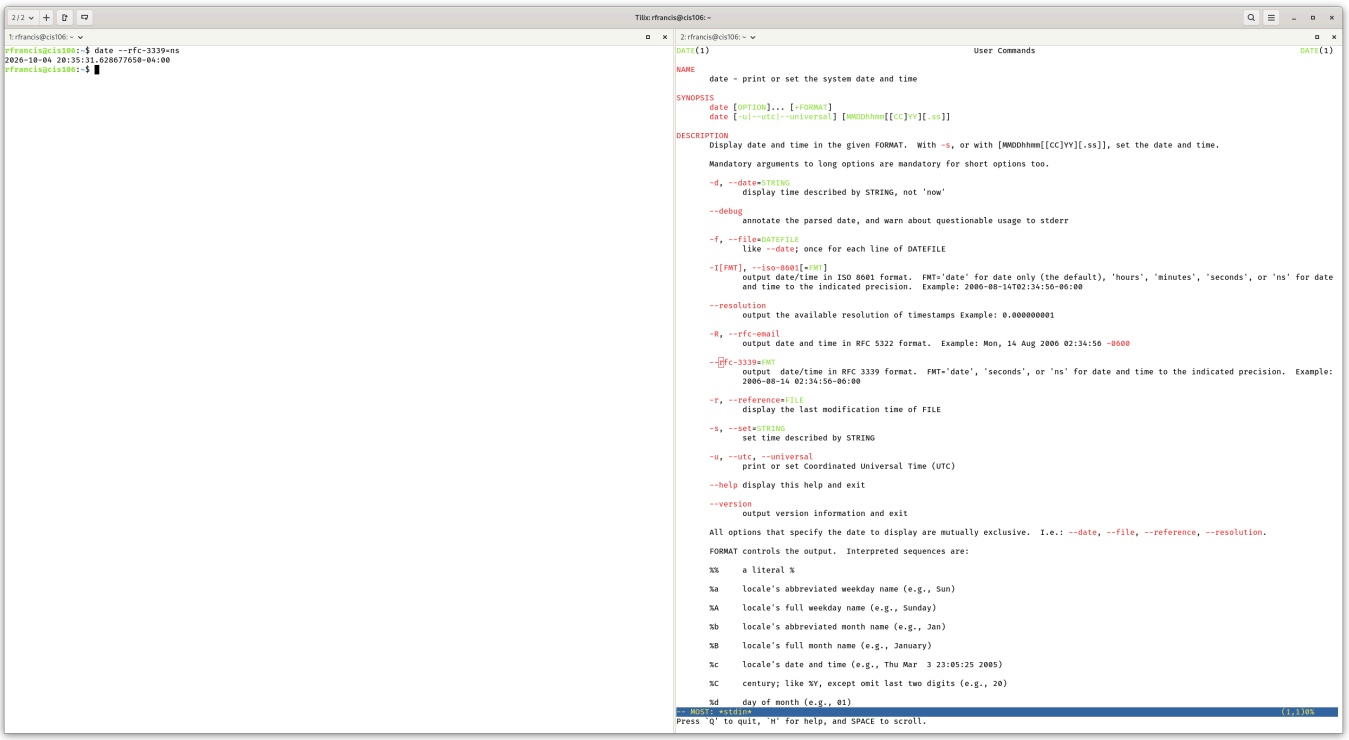
Question 3



Challenge Question


2 / 2
Tilix: rfrancis@cis106: ~
1: rfrancis@cis106: ~
2: rfrancis@cis106: ~
rfrancis@cis106:~$ free -h --line
SwapUse 2.1Gi CachUse 1.1Gi MemUse 2.9Gi MemFree 185Mi
rfrancis@cis106:~$
--kilo Display the amount of memory in kilobytes. Implies --si.
--mega Display the amount of memory in megabytes. Implies --si.
--giga Display the amount of memory in gigabytes. Implies --si.
--tera Display the amount of memory in terabytes. Implies --si.
--peta Display the amount of memory in petabytes. Implies --si.
-h, --human
Show all output fields automatically scaled to shortest three digit unit and display the units of print out. Following units are used.
B = bytes
Ki = kibibyte
Mi = mebibyte
Gi = gibibyte
Ti = tebibyte
Pi = pebibyte
If unit is missing, and you have exbibyte of RAM or swap, the number is in tebibytes and columns might not be aligned with header.
-w, --wide
Switch to the wide mode. The wide mode produces lines longer than 80 characters. In this mode buffers and cache are reported in two separate columns.
-c, --count count
Display the result count times. Requires the -s option.
-l, --lohi
Show detailed low and high memory statistics.
-L, --line
Show output on a single line, often used with the -s option to show memory statistics repeatedly.
-s, --seconds delay
Continuously display the result delay seconds apart. You may actually specify any floating point number for delay using either . or , for decimal point. usleep(3) is used for microsecond resolution delay times.
--si Use kilo, mega, giga etc (power of 1000) instead of kibi, mebi, gibi (power of 1024).
-t, --total
Display a line showing the column totals.
-v, --committed
Display a line showing the memory commit limit and amount of committed/uncommitted memory. The total column on this line will display the memory commit limit. This line is relevant if memory overcommit is disabled.
--help Print help.
-V, --version
Display version information.
FILES
/proc/meminfo
memory information
BUGS
The value for the shared column is not available from kernels before 2.6.32 and is displayed as zero.
Please send bug reports to
procps@freelists.org
SEE ALSO
ps(1), slabtop(1), top(1), vmstat(8).
procps-ng 2023-05-02 FREE(1)
-- MORE: *stdin* (53,1)100%
Press 'Q' to quit, 'H' for help, and SPACE to scroll.


2 / 2
Tilix: rfrancis@cis106: ~
1: rfrancis@cis106: ~
2: rfrancis@cis106: ~
rfrancis@cis106:~$ uname -srv -o
Linux 6.12.107+deb13-arm64 #1 SMP Debian 6.12.107-1 (2026-08-29) GNU/Linux
rfrancis@cis106:~$
UNAME(1) User Commands UNAME(1)
NAME
uname - print system information
SYNOPSIS
uname [OPTION]...
DESCRIPTION
Print certain system information. With no OPTION, same as -s.
-a, --all
print all information, in the following order, except omit -p and -i if unknown:
-s, --kernel-name
print the kernel name
-n, --nodename
print the network node hostname
-r, --kernel-release
print the kernel release
-v, --kernel-version
print the kernel version
-m, --machine
print the machine hardware name
-p, --processor
print the processor type (non-portable)
-i, --hardware-platform
print the hardware platform (non-portable)
-o, --operating-system
print the operating system
--help display this help and exit
--version
output version information and exit
AUTHOR
Written by David MacKenzie.
REPORTING BUGS
GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
Report any translation bugs to <https://translationproject.org/team/>
SEE ALSO
arch(1), uname(2)
Full documentation <https://www.gnu.org/software/coreutils/uname>
or available locally via: info '(coreutils) uname invocation'
Packaged by Debian (9.7-3)
Copyright © 2025 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
GNU coreutils 9.7 June 2025 UNAME(1)
-- MORE: *stdin* (1,1)100%
Press 'Q' to quit, 'H' for help, and SPACE to scroll.


2 / 2
Tilix: rfrancis@cis106: ~
1: rfrancis@cis106: ~
rfrancis@cis106:~$ date --rfc-3339=ns
2026-10-04 20:35:31.628677650-04:00
rfrancis@cis106:~$
2: rfrancis@cis106: ~
DATE(1) User Commands DATE(1)
NAME
date - print or set the system date and time
SYNOPSIS
date [OPTION]... [+FORMAT]
date [-u|--utc|--universal] [MMDDhhmm[[CC]YY][.ss]]
DESCRIPTION
Display date and time in the given FORMAT. With -s, or with [MMDDhhmm[[CC]YY][.ss]], set the date and time.
Mandatory arguments to long options are mandatory for short options too.
-d, --date=STRING
display time described by STRING, not 'now'
--debug
annotate the parsed date, and warn about questionable usage to stderr
-f, --file=DATEFILE
like --date; once for each line of DATEFILE
-I[FMT], --iso-8601[=FMT]
output date/time in ISO 8601 format. FMT='date' for date only (the default), 'hours', 'minutes', 'seconds', or 'ns' for date and time to the indicated precision. Example: 2006-08-14T02:34:56-06:00
--resolution
output the available resolution of timestamps Example: 0.000000001
-R, --rfc-email
output date and time in RFC 5322 format. Example: Mon, 14 Aug 2006 02:34:56 -0600
--rfc-3339=FMT
output date/time in RFC 3339 format. FMT='date', 'seconds', or 'ns' for date and time to the indicated precision. Example: 2006-08-14 02:34:56-06:00
-r, --reference=FILE
display the last modification time of FILE
-s, --set=STRING
set time described by STRING
-u, --utc, --universal
print or set Coordinated Universal Time (UTC)
--help display this help and exit
--version
output version information and exit
All options that specify the date to display are mutually exclusive. I.e.: --date, --file, --reference, --resolution.
FORMAT controls the output. Interpreted sequences are:
%% a literal %
%a locale's abbreviated weekday name (e.g., Sun)
%A locale's full weekday name (e.g., Sunday)
%b locale's abbreviated month name (e.g., Jan)
%B locale's full month name (e.g., January)
%c locale's date and time (e.g., Thu Mar 3 23:05:25 2005)
%C century; like %Y, except omit last two digits (e.g., 20)
%d day of month (e.g., 01)
(1,1)0%
Press 'Q' to quit, 'H' for help, and SPACE to scroll.